



Bukku Coding Assignment – Laravel (Test 1)

Problem Statement

One of your friends running an e-commerce business is constantly having problems with tracking the cost of the goods that they are selling online accurately.

He came to you asking if you can build him a backend system for him to track their purchase and sale records, and the system should be capable of recording the cost for each of the sale records.

He also shared with you that it's important that he would like the costing to be based on Weighted Average Cost (WAC) method.

Weighted Average Cost (WAC)

WAC uses the average cost of all items you are holding at a given point of time to determine the cost for each sale you made.

For example:

- You bought 150 units of a product at RM2 each on 01/01/2022, the total value of goods that you're holding on hand will be $150 \times \text{RM}2 = \text{RM}300$.
- The average cost of the product that you're holding on/after the date will be RM2.
- You bought another 10 units at RM1.50 each on 05/01/2022, the total value of goods on hand you have added is RM15.
- The total number of units you are holding on/after that day will be 160, the total value will be RM315, the average cost will be $\text{RM}315 / 160 = \text{RM}1.97$ (if rounded to 2 decimal points)
- If you were to sell 5 units afterwards, say on 07/01/2022 at whatever price to your customer, the following will happen:
 - The cost for the items sold will be $\text{RM}1.97 \times 5 = \text{RM}9.85$, which means the total value of goods will be reduced by RM9.85, becoming RM305.15 and total quantity on hand will be 155.

The Task

Build a RESTful application with Laravel (or any other modern web framework) to track purchase and sale transactions, at the same time calculate the cost for each sale recorded.

The system should have:

- Basic user register / login capabilities powered by JWT authentication.
- A way to record new purchase transaction.
- A way to retrieve a list of purchase transactions.
- A way to record new sale transaction.
- A way to retrieve a list of sale transactions, with costing information.

To keep it straightforward:

- There can only be one transaction on each date, either sale OR purchase.
- Transactions cannot be updated or deleted.



- Creation of new transaction can only be following date sequence, oldest to latest.
- You may seed a list of dummy products.
- Each sale or purchase only deals with one single product instead of multiple products, however, take note of the support for quantity at transaction level.
- Only RESTful endpoints are required, no frontend / user interface is required.

Bonus (Try if you're adventurous):

1. Allowing creation of transactions in random date order and system to automatically adjust other transactions accordingly (efficiently).
2. Allowing updating / deleting existing transactions.

Evaluation Criteria

- Your understanding of the problem statement.
- Making reasonable assumptions.
- Proper documentation on how to navigate / use the system.
- Reasonable RESTful endpoints naming convention.
- Proper validation rules.
- Proper test cases.
- Code quality and conciseness with proper comments.

Submission

- Push to your GitHub account and share the GitHub link to your point of contact at Bukku.